



几何测度论与偏微分方程 研讨会



国家天元数学中部中心
武汉大学数学与统计学院
国家天元数学中部中心恩施基地
湖北民族大学数学与统计学院

二〇二六年八月十六日至二十二日

意大利 Bocconi 大学

摘要: In this talk, I will discuss a Simon-Smith min-max theory for anisotropic minimal surfaces in closed three-manifolds, with controlled genus. The starting point is an anisotropic analogue of the celebrated theorem of Meeks–Simon–Yau: every minimizing sequence of surfaces within a fixed isotopy class converges, as varifolds, to a smooth stable anisotropic minimal surface, possibly with multiplicity, with lower semicontinuity of genus. This result strengthens previous existence theorems for anisotropic Plateau problems and provides the compactness needed for a topological min-max construction. As an application, we prove the existence of closed embedded anisotropic minimal surfaces with controlled genus in arbitrary closed three-manifolds. This is joint work with Antonio De Rosa and Aria Halavati.

Singular Limits for Three-Dimensional Global Minimizers of Ginzburg-Landau-Type Functionals: Uniform Estimates and Singular Sets

王尉

北京大学

摘要: We investigate the asymptotic behavior of global minimizers of a Ginzburg-Landau-type functional with a general compact vacuum manifold on bounded domains in three-dimensional Euclidean space, in the regime where the energy grows logarithmically. We prove that, along a subsequence, the normalized energy measures converge to a measure supported on a finite union of closed line segments connecting prescribed boundary singularities. We also analyze the associated singular sets and establish uniform estimates for the model.

Asymptotic limit of isotropic-nematic phase transition for liquid crystals with anisotropic elasticity

王

大学